



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 04.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : PHA G 612
Course Title : Pharmacokinetics & Clinical Pharmacy
Instructor-in-Charge : Dr. Pragyanshu Khare
Instructor(s) : NA
Tutorial/Practical Instructors: Jagdish Chand, Shivangi Paliwal, Shivangi Nema, Dhruv Sharma, Saral Bachuka

1. Course Description:

Introduction to pharmacokinetic modeling of drugs; Compartmental modeling including one compartmental and two compartmental models; Non-compartmental analysis (NCA); Bioequivalence and comparative bioavailability studies; Pharmacokinetic Pharmacodynamic modeling techniques, determination of order of absorption and absorption rate constant by Wagner-Nelson method; non-linear pharmacokinetics; Pharmacokinetic drug interactions; multi-dose pharmacokinetics of drugs following one compartmental model; Design of multi-dose regimen; Dosage regimen adjustment in renal impairment; Application of the pharmacokinetic principles for better therapeutic outcomes

2. Scope and Objective of the Course:

The course imparts knowledge of pharmacokinetics, its application, pharmacokinetic studies, mathematical modeling, and experimental design. This will help students to gain knowledge in a very important area of pharmacy. Practicals are designed to expose students to pharmacokinetic studies.

3. Text Books:

- i. T1. Gibaldi, M & Perrier, O., Pharmacokinetics, Marcel Dekker, N.Y., 2nd ed. 1993.

4. Reference Books:

- i. Herfindal and Hirschman., Clinical Pharmacy and Therapeutics.
- ii. Remington., Pharmaceutical Sciences.
- iii. Curr and Whelpton., Pharmacokinetics Lab. Manual.
- iv. Gibaldi M. & Perrier O., Pharmacokinetics, Marcel Dekker, N Y, 1st Edn.
- v. Gibaldi M., Biopharmaceutics and Clinical Pharmacokinetics, 4th Edn.
- vi. Journals.

5. Course Plan:

Weeks	Module Number	Learning objective	Reference	Learning Outcome
1-3	1. Introduction and overview of Pharmacokinetics	L1.1 Introduction to Pharmacokinetics and its Application	R 1, R2	Knowledge of Pharmacokinetics & ADME characters of drugs, Compartment model: one, two & multi-compartments
		L1.2 Concept of compartment in pharmacokinetics	R1, R7	



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4-8	2. Pharmacokinetics (ADME) studies after different routes of administration and Compartment Models	L2.1 Pharmacokinetic study of drugs and metabolites: in plasma & urine following i.v bolus	T1-1	Knowledge of one compartment i.v. bolus i.v. infusion, 1st order absorption & zero order absorption process and estimation of PK parameters. Student will acquire knowledge on two and multicompartment models.
		L2.2 Pharmacokinetics of drugs by iv infusion	T1-1	
		L2..3 Pharmacokinetics of drugs by IV Bolus; 2- compartment model		
		L2.3 Pharmacokinetics of drugs administered through oral administration	T1-1	
9-12	3. Multiple Dosing Studies and Non Compartment Analysis	L3.1 Pharmacokinetics of one compartment multiple dosing : drug in plasma, iv injection, 1st order absorption	T1-2	In depth knowledge of multiple dosing and non-compartmental analysis
		L3.1 Pharmacokinetics of one compartment multiple dosing : drug in plasma, oral administration, 1st order	T1-3	
13-15	4. Bioavailability, Bioequivalence studies, physiological and pharmacokinetic models	4.1 Bioavailability: Rate and extent of absorption	T1-4	Study of bioavailability and bioequivalence, In-depth knowledge of various PK parameters and PBPK studies
		4.2 Study of nonlinear pharmacokinetics	T1-7	
		4.3 Concept and study of apparent volume of distribution	T1-5	
16-18	5. Pharmacokinetic parameters in hepatic and renal impairment, Q.S.Pk.R, Experimental techniques in pharmacokinetics	L5.1 Various applications of pharmacokinetic studies ; including in renal impairment	T1-10	Application of pharmacokinetic Studies, Quantitative structure Pharmacokinetic
19-20	6. PK-PD relationship : determination and predictions	6.1 PK-PD relationship : determination and predictions	Literature, Journals and notes	Understanding of Population Pharmacokinetics PK-PD relationship



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6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Close Book/ Open Book
Mid-Semester Test	90 Min.	30	TBD	CB/OB
Comprehensive Examination	180 Min	35	TBD	CB/OB
Continuous Assessment*	TBD	25	TBD	CB/OB
Class discussions* (Linked to attendance)	In class	10	In class	NA

*Quiz(es) will be conducted as a part of the evaluation component, at random as convenient with/without prior intimation, and hence it is expected that the students come prepared to every class on topics covered in earlier contact hours.

Criterion for NC: 30% of the average of the top three performers or 40% of median, whichever is lower, or a suitable modification at the discretion of the IC with the concurrence of the Department

7. Consultation Hour: To be announced in the class.

8. Notices: All the notices pertaining to this course will be displayed on Nalanda.

9. Make-up Policy: Make-ups are not given as a routine. It solely depends on the “genuineness” of the circumstances under which a student fails to appear in a scheduled evaluation component. Prior permission should be sought from the instructor-in-charge in advance.

All evaluation components are equally important, irrespective of weightage. Hence, students failing to attend scheduled classes, or absenting themselves in one or many of the evaluation components, may become ineligible for obtaining a valid grade at the end of the semester.

Hence, students are strongly advised to keep away from absenting themselves from all aforementioned contact sessions. Clearing the course would require adequate performance in quizzes**, tests**, examinations**, and in practical (laboratory) components, separately

Instructor-in-charge
Course No. PHA G612